Mass Spring System Undamped Motion

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5.1.1 SPRING/MASS SYSTEMS: FREE UNDAMPED MOTION

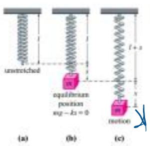

FIGURE 5.1.1 Spring/mass system

Hooke's Law Suppose that a flexible spring is suspended vertically from a rigid support and then a mass m is attached to its free end. The amount of stretch, or elongation, of the spring will of course depend on the mass; masses with different weights stretch the spring by differing amounts. By Hooke's law the spring itself exerts a restoring force F opposite to the direction of elongation and proportional to the amount of elongation s. Simply stated, $F = ks$, where k is a constant of proportionality called the **spring constant.** The spring is essentially characterized by the number k. For example, if a mass weighing 10 pounds stretches a spring $\frac{1}{2}$ foot, then $10 = k\left(\frac{1}{2}\right)$ implies $k = 20$ lb/ft. Necessarily then, a mass weighing, say, 8 pounds stretches the same spring only $\frac{2}{5}$ foot.

Newton's Second Law After a mass m is attached to a spring, it stretches the spring by an amount s and attains a position of equilibrium at which its weight W is balanced by the restoring force ks. Recall that weight is defined by $W = mg$, where mass is measured in slugs, kilograms, or grams and $g = 32$ ft/s^2, 9.8 m/s^2, or 980 cm/s^2, respectively. As indicated in Figure 5.1.1(b), the condition of equilibrium is $mg = ks$ or $mg - ks = 0$. If the mass is displaced by an amount x from its equilibrium position, the restoring force of the spring is then $k(x + s)$. Assuming that there are no retarding forces acting on the system and assuming that the mass vibrates free of other external forces—**free motion**—we can equate Newton's second law with the net, or resultant, force of the restoring force and the weight:

$$m\frac{d^2x}{dt^2} = -k(s + x) + mg = -kx + \underbrace{mg - ks}_{\text{zero}} = -kx. \qquad (1)$$

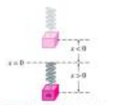

FIGURE 5.1.2 Direction below the equilibrium position is positive.

The negative sign in (1) indicates that the restoring force of the spring acts opposite to the direction of motion. Furthermore, we adopt the convention that displacements measured *below* the equilibrium position $x = 0$ are positive. See Figure 5.1.2.

DE of Free Undamped Motion By dividing (1) by the mass m, we obtain the second-order differential equation $d^2x/dt^2 + (k/m)x = 0$, or

$$\frac{d^2x}{dt^2} + \omega^2 x = 0, \qquad (2)$$

where $\omega^2 = k/m$. Equation (2) is said to describe **simple harmonic motion** or **free undamped motion.** Two obvious initial conditions associated with (2) are $x(0) = x_0$ and $x'(0) = x_1$, the initial displacement and initial velocity of the mass, respectively. For example, if $x_0 > 0$, $x_1 < 0$, the mass starts from a point *below* the equilibrium position with an imparted *upward* velocity. When $x'(0) = 0$, the mass is said to be released from rest. For example, if $x_0 < 0$, $x_1 = 0$, the mass is released from *rest* from a point $|x_0|$ units *above* the equilibrium position.

$$m\frac{d^2x}{dt^2} + kx = 0$$

$$\frac{d^2x}{dt^2} + \left(\frac{k}{m}\right)x = 0$$

$$\frac{d^2x}{dt^2} + \omega^2 x = 0$$

$$m = \frac{w}{g}$$

$$F = kx.$$

EXAMPLE 1 Free Undamped Motion

A mass weighing 2 pounds stretches a spring 6 inches. At $t = 0$ the mass is released from a point 8 inches below the equilibrium position with an upward velocity of $\frac{4}{3}$ ft/s. Determine the equation of motion.

Solⁿ:

$F = 2\,lb$

$x = 6$ inch $= \frac{1}{2}$ ft

$F = kx$

$\Rightarrow 2 = k\left(\frac{1}{2}\right)$

$\Rightarrow k = 4\ lb\,ft^{-1}$

$m = \frac{2}{32} = \frac{1}{16}$ slug (Divide by 32 to get slug from pound)

$\therefore \omega = \sqrt{\frac{k}{m}}$

$= 8$

$\therefore \frac{d^2x}{dt^2} + 64x = 0$ ——— (i)

Solving (i) $\Rightarrow x(t) = C_1\cos 8t + C_2\sin 8t$

Given conditions, $x(0) = 8$ inches $= \frac{2}{3}$ ft

$x'(0) = -\frac{4}{3}\,\text{fts}^{-1}$ (-ve because upward velocity)

Now, $\frac{2}{3} = C_1 \cos 0 + C_2 \sin 0 \Rightarrow C_1 = \frac{2}{3}$

$x'(t) = -8C_1 \sin 8t + 8C_2 \cos 8t$

$\Rightarrow -\frac{4}{3} = -8C_1 \sin 0 + 8C_2 \cos 0$

$\Rightarrow C_2 = -\frac{1}{6}$

$\therefore\ x = \frac{2}{3}\cos 8t - \frac{1}{6}\sin 8t.$

11. A mass weighing 64 pounds stretches a spring 0.32 foot. The mass is initially released from a point 8 inches above the equilibrium position with a downward velocity of 5 ft/s.

(a) Find the equation of motion.

(b) What are the amplitude and period of motion?

(c) How many complete cycles will the mass have completed at the end of 3π seconds?

(d) At what time does the mass pass through the equilibrium position heading downward for the second time?

(e) At what times does the mass attain its extreme displacements on either side of the equilibrium position?

(f) What is the position of the mass at $t = 3$ s?

(g) What is the instantaneous velocity at $t = 3$ s?

(h) What is the acceleration at $t = 3$ s?

(i) What is the instantaneous velocity at the times when the mass passes through the equilibrium position?

(j) At what times is the mass 5 inches below the equilibrium position?

(k) At what times is the mass 5 inches below the equilibrium position heading in the upward direction?

11)

$F = 64\,\text{lb}$

$x = 0.32\,\text{ft} =$

$x(0) = -8$ inches $= -\frac{2}{3}$ ft

$x'(0) = 5\,\text{fts}^{-1}$

a) $F = kx$

$\Rightarrow 64 = k \cdot 0.32$

$\Rightarrow k = 200 \text{ lb ft}^{-1}$

$\therefore m = \dfrac{64}{32} = 2 \text{ slug}$

$\therefore \omega = \sqrt{\dfrac{k}{m}} = \sqrt{\dfrac{200}{2}} = \sqrt{100} = 10$

$\therefore \dfrac{d^2x}{dt^2} + 100x = 0$

$\hookrightarrow$ solution of this equation: $x(t) = C_1 \cos 10t + C_2 \sin 10t$

$\Rightarrow -\dfrac{2}{3} = C_1 \cos 0 + C_2 \sin 0$

$\Rightarrow C_1 = -\dfrac{2}{3}$

Then, $x'(t) = -10 C_1 \sin 10t + 10 C_2 \cos 10t$

$\Rightarrow 5 = -10 \times C_1 \sin 0 + 10 C_2 \cos 0$

$\Rightarrow C_2 = \dfrac{1}{2}$.

$\therefore x(t) = -\dfrac{2}{3} \cos 10t + \dfrac{1}{2} \sin 10t$.

b) Amplitude, $A = \sqrt{C_1^2 + C_2^2} = \dfrac{5}{6}$

Period of motion, $T = \dfrac{2\pi}{\omega} = \dfrac{2\pi}{10} = \dfrac{1}{5}\pi$

c) By T seconds, the mass will have completed one cycle

By $\frac{2}{5}\pi$ " " " " " " 1 "

By 3π " " " " " " 15 cycles

d) At equilibrium, $x=0$

Here, phase angle, $\varphi = \tan^{-1}\frac{c_1}{c_2} = \tan^{-1}\left(-\frac{4}{3}\right)$

$\therefore x(t) = A\sin(\omega t + \varphi)$

$0 = \frac{5}{6}\sin\left(10t + \tan^{-1}\left(-\frac{4}{3}\right)\right)$

$\Rightarrow 10t + \tan^{-1}\left(-\frac{4}{3}\right) = n\pi$

$\Rightarrow 10t + \tan^{-1}\left(-\frac{4}{3}\right) = 2\pi$

$\Rightarrow t = 0.72104$ sec.

$\sin x = 0$

$\Rightarrow x = n\pi$

Here, $n=2$.

e) $x(t) = \frac{5}{6}\sin\left(10t + \tan^{-1}\left(-\frac{4}{3}\right)\right)$

$\Rightarrow x'(t) = \frac{25}{3}\cos\left(10t + \tan^{-1}\left(-\frac{4}{3}\right)\right)$

Extreme displacement is attained when, $x'(t) = 0$

$\Rightarrow \cos\left(10t + \tan^{-1}\left(-\frac{4}{3}\right)\right) = 0$

$\Rightarrow 10t + \tan^{-1}\left(-\frac{4}{3}\right) = (2n+1)\frac{\pi}{2}$

$\Rightarrow t = (2n+1)\frac{\pi}{20} - \frac{1}{10}\tan^{-1}\left(-\frac{4}{3}\right)$

$\cos x = 0$

$x = (2n+1)\frac{\pi}{2}$

$\therefore t = (2n+1)\frac{\pi}{20} - \frac{1}{10}\tan^{-1}\left(-\frac{4}{3}\right)$ where, $n = 0, 1, 2, 3, 4 \ldots$

f) $x(3) = \frac{5}{6}\sin\left(10\times 3 + \tan^{-1}\left(-\frac{4}{3}\right)\right)$

$= -0.597$ ft

g) $x'(3) = \frac{25}{3}\cos\left(10\times 3 + \tan^{-1}\left(-\frac{4}{3}\right)\right)$

$= -5.8156$ ft/sec

h) $x''(t) = -\frac{250}{3}\sin\left(10t + \tan^{-1}\left(-\frac{4}{3}\right)\right)$

$\therefore x''(3) = 59.685\ \text{ft}^2/\text{sec}$

i) When passes through the equilibrium position,

$$x(t) = 0$$

$$\Rightarrow \frac{5}{6}\sin\left(10t + \tan^{-1}\left(-\frac{4}{3}\right)\right) = 0$$

$$\Rightarrow t = \frac{n}{10}\pi - \frac{1}{10}\tan^{-1}\frac{4}{3}$$

Now, $x'(t) = \frac{25}{3}\cos\left(10t + \tan^{-1}\left(-\frac{4}{3}\right)\right)$

$$= \frac{25}{3}\cos(n\pi)$$

$$= \frac{25}{3} = 8.33 \text{ ft sec}^{-1}.$$

j) $x(t) = \frac{5}{12}$ ft $\left(5 \text{ inches} = \frac{5}{12} \text{ ft}\right)$

$$\Rightarrow \frac{5}{6}\sin\left(10t + \tan^{-1}\left(-\frac{4}{3}\right)\right) = \frac{5}{12}$$

$$\Rightarrow 10t + \tan^{-1}\left(-\frac{4}{3}\right) = n\pi + (-1)^n\frac{\pi}{6}$$

$$\Rightarrow t = \frac{n}{10}\pi + (-1)^n\frac{\pi}{60} - \frac{1}{10}\tan^{-1}\left(-\frac{4}{3}\right)$$

$\sin x = \sin\theta$

$x = n\pi + (-1)^n\theta$

Here, $\theta = \frac{\pi}{6}$.

where, $n = 0, 1, 2, 3, 4, 5, \ldots$

1.1 same as j

k) same as j